

31. As we are in a 2-D plane, the linear classifier here will be a straight line separating 0s from 1s.

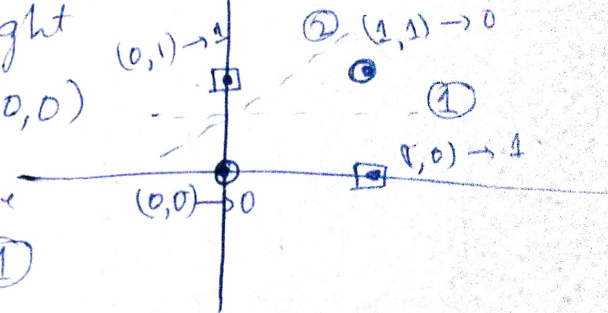
From the diagram we can clearly

see that if there is any straight line which separates $(0,1)$ from $(0,0)$

will necessarily place $(1,1)$ on the

same side as $(0,1)$ as in Line ①

or it will not keep $(1,0)$ on the same side as $(0,1)$ as in line ②.



Therefore there does not exist any decision boundary which can separate the XOR classes. and hence XOR is not linearly separable.